Unit Examination 1

Due: Wednesday, October 7 at 9:00 AM **Value: 125 points** **Format: 50-minute in-class examination**

Purpose

Apply core ideas about macromolecules, membranes, organelles, microscopy, and experimental controls.

Your task

- Part A (25): Answer five 5-point mechanism questions: (1) Why can a hydrophobic substitution disrupt a protein core? (2) Why does carrier-mediated uptake saturate? (3) What distinguishes an isotonic solution from an iso-osmotic but penetrating solution? (4) Why does a lysosomal enzyme require targeting information? (5) How can a cytoskeletal motor move directionally without sensing its destination?
- Part B1 (10): From the glucose-uptake table, estimate the maximum observed rate and the concentration that first reaches at least half that rate.
- Part B2 (10): Decide whether the pattern is more consistent with simple diffusion or carrier-mediated transport and cite two observations.
- Part B3 (10): Predict the curve after halving transporter abundance and distinguish rate capacity from equilibrium.
- Part C (30): Trace a secreted peptide from translation initiation to release outside the cell. Then predict the protein's location and modification after (a) loss of the ER signal sequence and (b) Golgi vesicle-formation failure.
- Part D (25): A fluorescent drug appears to move a receptor from membrane to lysosome. Design vehicle, no-primary-antibody, known-lysosome-marker, and untreated controls; state what result each control validates.
- Part E (15): A 1.0 mm stage-micrometer span covers 40 ocular divisions. A cell spans 6.5 divisions. Calculate its size and draw the printed length of a 20-micrometer scale bar when 100 micrometers prints as 25 mm.

Provided evidence or data

External glucose	Initial uptake rate
1 mM	2 units/min
5 mM	8 units/min
10 mM	12 units/min
50 mM	14 units/min

Submit

- Completed examination booklet and approved handwritten reference card.

Important note

Closed note except for one handwritten 4 x 6 inch reference card. Basic calculators are permitted.

Grading criteria

Criterion	Pts	Full-credit evidence
Part A	25	Selects and justifies the correct mechanism.
Part B	30	Reads the graph accurately and explains transport behavior.
Part C	30	Correct pathway, compartments, and disruption predictions.
Part D	25	Controls isolate the intended causal factor.
Part E	15	Correct conversions, units, and scale bar.